HARF (حرف) – Project Planning & Management

Interactive Speech Practice Platform for Children | DEPI Capstone Project | React Full-Stack Track

Team: Team Leader / Frontend (Ahmed – Backend, Randa – UI/UX, Mariam – Database, Hager – Testing & AI Integration)

1. Project Proposal

1.1 Overview

HARF is a child-friendly, game-based web platform that makes daily speech practice fun and consistent. It focuses on **pronunciation (articulation) practice** for children. It gives parents and speech-language specialists the tools to follow and guide that practice. HARF supports the specialist and never replaces them.

1.2 Problem Statement

Speech therapy sessions are short and infrequent, so most practice has to happen at home. Parents often lack structured material, children lose interest in repetitive drills, and specialists cannot easily see what happened between sessions. Quality Arabic-friendly digital tools for this purpose are still limited.

1.3 Proposed Solution

One connected platform for the child, the parent, the specialist, and the admin, with interactive pronunciation activities, levels and stages, gamification, personalized training plans, and progress reports.

1.4 Objectives

- Make daily pronunciation practice engaging through games, points, levels, and badges.
- Let parents create and manage a profile for their child and follow progress.
- Let specialists build personalized training plans, assign activities, and monitor results.
- Provide clear progress tracking between therapy sessions.

- Use AI (prompt engineering) as a supporting tool for specialists to suggest and draft activities and summarize progress. It is never used for diagnosis.

1.5 Target Users

User	Description
Child	Uses the platform to play and practice pronunciation. The account is created and managed by the parent.
Parent / Guardian	Creates the child's account, follows progress, receives notifications, and subscribes the child to a specialist if desired.
Specialist	Manages assigned children, builds training plans, assigns activities, and reviews reports.
Admin	Manages users, roles, activities, and content, and monitors the platform.

1.6 Usage Modes

Mode	Description
Free Interactive Mode	Open to any child. The child progresses through levels and stages of pronunciation games and earns points, badges, and achievements.
Specialist Subscription Mode	The parent links the child to a specialist through a subscription package (request and approval). The specialist then builds a personalized plan and follows the child's results, with face-to-face sessions arranged between them. Payment processing is not included in the project scope.

1.7 Scope

In scope

- Pronunciation-focused games (letters, sounds, words) with levels and stages.

- Role-based accounts: Parent, Child, Specialist, Admin, with JWT authentication.
- Gamification: points, levels, badges, challenges.
- Specialist tools: training plans, activity assignment, statistics, and reports.
- Parent dashboard: progress, assigned and completed activities, notifications.
- AI-assisted activity suggestions and progress insights for specialists.
- Admin panel for users and content.

Out of scope

- Online payment and billing.
- Medical diagnosis or any replacement of the specialist.
- General language and comprehension therapy (vocabulary, grammar, sentence building).
- Automatic speech recognition and scoring of the child's voice.
- Built-in video calling (sessions with the specialist happen outside the platform).

1.8 Technology Stack

Layer	Tools
Frontend	React.js, TypeScript, React Router, Tailwind CSS, Redux Toolkit, Canvas/SVG and CSS animations for games and rewards
Backend	Node.js, Express.js, RESTful APIs, JWT with role-based authorization
Database	MongoDB
AI	Prompt-engineering integration for specialist support
Testing	Jest and React Testing Library
DevOps and Collaboration	Docker and Docker Compose, Git and GitHub (branches, pull requests, code reviews)
Design	Figma (personas, wireframes, accessibility)

2. Project Plan

2.1 Timeline (Gantt Chart)

Phase	Oct 15–16	Oct 17–Oct 23	Oct 24–Nov 6	Nov 7–Nov 30	Dec 1–Dec 4
Planning and Management	■■■				
Literature Review	■■■				
Requirements Gathering	■■■	■			
System Analysis and Design		■■■	■■■		
Implementation (Frontend, Backend, DB)			■■	■■■■■■	
Testing, Reports, and Final Presentation				■■	■■■

This timeline is a draft based on the program deadlines. Adjust it to your real dates if needed.

2.2 Milestones

Date	Milestone
16 Oct 2026	Project Planning and Management, Literature Review, and Requirements Gathering submitted
6 Nov 2026	System Analysis and Design submitted (wireframes, database schema, API design)
30 Nov 2026	Implementation completed (source code and execution)
4 Dec 2026	Testing, Reports, and Final Presentation delivered

2.3 Deliverables

Phase	Deliverables
Planning	Proposal, project plan, roles, risk plan, KPIs

Phase	Deliverables
Requirements	User personas, user stories, acceptance criteria
Design	Wireframes and UI design (Figma), database schema, API specification, system workflows
Implementation	Frontend (React), Backend (Express APIs), MongoDB database, AI integration, Docker setup
Testing and Reports	Jest and React Testing Library tests, test report, final report, presentation

2.4 Resource Allocation

Resource	Usage
GitHub	Source control with branches, pull requests, and code reviews
Figma	UI/UX design and wireframes
Postman	API testing and documentation
MongoDB Atlas or local MongoDB	Database
Docker and Docker Compose	Running frontend, backend, and database together
WhatsApp and Google Meet	Team communication and meetings

3. Task Assignment and Roles

Member	Role	Responsibilities
Team Leader	Team Leader and Frontend Developer	Plans and coordinates the team, builds the React frontend (Tailwind, Redux Toolkit, Router), integrates the frontend with the APIs, and prepares planning documents

Member	Role	Responsibilities
Ahmed	Backend Developer	Builds Node.js and Express REST APIs, JWT authentication and role-based authorization, and API documentation
Randa	UI/UX Designer	Creates personas, wireframes, and Figma designs, and designs the child-friendly game visuals and accessibility
Mariam	Database Engineer	Designs the MongoDB schemas (users, children, activities, plans, progress, achievements, notifications) and supports the backend with queries and data seeding
Hager	QA Engineer, AI Integration, and DevOps	Writes Jest and React Testing Library tests, prepares prompts for the AI features, and sets up Docker and Docker Compose

4. Risk Assessment and Mitigation Plan

Risk	Probability	Impact	Mitigation
Limited time before the final deadline	High	High	Focus on a Minimum Viable Product first, and keep extra features optional
Scope grows beyond what the team can build	Medium	High	Keep to the agreed scope and focus only on pronunciation
Merge conflicts in the code	Medium	Medium	Separate branches per member, pull requests, and code reviews
Delay in one member's work blocks others	Medium	High	Agree on API contracts early, use mock data in the frontend, and hold short regular check-ins
Child data privacy and safety	Medium	High	Parent-managed accounts, JWT authentication, hashed passwords, and role-based access

Risk	Probability	Impact	Mitigation
AI produces unsuitable suggestions	Medium	Medium	Specialist always reviews and approves, and the AI is never used for diagnosis
Technical problems during the final demo	Low	High	Docker setup, a prepared demo account with sample data, and a rehearsal before the presentation
A team member becomes unavailable	Low	Medium	Documented work, shared repository, and backup task owners

5. KPIs (Key Performance Indicators)

KPI	Target
Response time of main pages and API calls	Under 2 seconds
System uptime during testing and demo	99% or higher
Core features completed (authentication, games, progress tracking, specialist plans)	100% of the agreed MVP
Automated test coverage of core components and user flows	At least 70%
Activity completion rate in user testing	80% or higher
User satisfaction from a small test group (parents and children)	4 out of 5 or higher
Responsiveness	Works correctly on mobile, tablet, and desktop
Team delivery	All phase deadlines met